



AI humanizers and the crisis of information integrity: implications for scientific writing

Louie Giray^{1,2}

Received: 24 February 2026 / Accepted: 6 March 2026 / Published online: 18 March 2026

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026, modified publication 2026

Abstract

AI humanizers are tools designed to disguise AI-generated text as human-written. These tools have triggered a crisis in information integrity. This paper examines the reasons behind the rise of AI humanizers and their damaging effects on scientific writing. The primary reasons for adopting AI humanizers stem from institutional and psychological pressures. Students and researchers face a *publish-or-perish* culture and rigid academic policies that impose arbitrary AI-percentage thresholds. These tools are increasingly used as compliance mechanisms to avoid being flagged by AI detection. Furthermore, a lack of confidence in information literacy and the pressure of time poverty drive users to seek an information bypass, prioritizing efficient yet shallow output over deep cognitive engagement. The effects of this reliance are ethically and educationally corrosive. By decoupling writing from thinking, AI humanizers produce *pseudo-information* that mimics scholarly fluency but lacks epistemic grounding. This leads to a decline in critical information processing. Long-term consequences include the stunting of information literacy, the rise of learned helplessness, and the systemic pollution of institutional repositories with content lacking authentic inquiry. The paper argues that information integrity cannot be recovered through technical policing alone; it requires a return to process-oriented assessments that prioritize information provenance and the authentic knowledge engagement.

Keywords AI humanizers · Information ethics · Information integrity · Information literacy · Pseudo-knowledge · Scientific writing

Introduction

What happens to information when its origin becomes unknowable? Consider an academic paper submitted by a student. The language appears human, as verified by AI detection software. The citations seem legitimate, although closer inspection reveals that some links do not function. The argument is coherent and robotically structured. Yet the text did not originate from human cognition. Instead, it was generated by AI and subsequently rewritten by an AI humanizer to evade AI detection. This scenario raises fundamental epistemological questions. What is the status of such information? Can it be considered knowledge, and

does it meaningfully contribute to the institutional repository of human understanding?

These questions strike at the core of information studies in the age of AI. Since the public release of ChatGPT in late 2022, academic information production has experienced an unprecedented disruption. AI can now generate seemingly complete research papers in a matter of minutes (Baker et al. 2023), and early cases have documented AI tools being listed as co-authors in scholarly publications (Stokel-Walker 2023).

Crucially, this expansion of generative capacity has been accompanied by the emergence of AI humanizers. While generative AI produces academic-style text, AI humanizers are explicitly designed to rewrite such outputs to bypass AI detection. Their purpose is concealment. By introducing controlled imperfections and stylistic variability, AI humanizers obscure the provenance of information and sever the traceability that academic institutions rely upon to establish authenticity and credibility (Weber-Wulff et al. 2023).

Recent reporting highlights widespread student use of AI humanizer to evade detection, raising significant concerns

✉ Louie Giray
lggiray@mapua.edu.ph

¹ Department of Liberal Arts, School of Foundational Studies and Education, Mapua University, Manila, Philippines

² Research Center for International and Global Higher Education, Khazar University, Baku, Azerbaijan

for academic integrity and fairness (Daniel 2025; Rabino-wicz 2025). Google Trends data further suggests that this practice is not isolated, with particularly high search concen-trations in the Philippines, indicating a systemic information problem rather than individual misconduct.

Theoretical frameworks

This paper examines AI humanizers through three frameworks: (1) the DIKW hierarchy, (2) the information behavior theory, and (3) the philosophy of information. It explores how AI humanizers disrupt information flows and challenge assump-tions about what counts as legitimate academic information.

Data-Information-Knowledge-Wisdom (DIKW) hierarchy

The Data-Information-Knowledge-Wisdom (DIKW) hierarchy has long served as a model in information studies for under-standing how raw data transforms into actionable wisdom. In this model, data represents raw, unprocessed facts; informa-tion is data given context and meaning; knowledge emerges when information is internalized, understood, and connected to existing mental models; and wisdom represents the applied, ethical use of knowledge in decision-making (Rowley 2007). AI humanizers fundamentally disrupt this hierarchy. When users utilize these tools, they skip from data directly to what *appears* to be information. This bypasses the cogni-tive processes that transform data into genuine knowledge. The output may look like information: it has structure, cita-tions, coherent arguments, but it lacks the authentic connec-tion between information content and the human cognitive processes that validate and contextualize it (see Table 1).

Philosophy of information

The philosophy of information, particularly as articulated by Luciano Floridi (2013), emphasizes that information is not merely content; it carries with it questions of truth, meaning, and origin. Floridi’s principle of information ethics holds that information entities have inherent value and dignity,

and that tampering with information integrity constitutes an ethical violation (Floridi 2013).

AI humanizers violate this principle in multiple ways:

1. **Source falsification.** They disguise AI-generated infor-mation as human-generated, breaking information prov-eance.
2. **Semantic manipulation.** They alter text specifically to deceive verification systems.
3. **Epistemic deception.** They present non-knowledge as knowledge, corrupting institutional information systems.

Information behavior theory

Information behavior theory examines how individuals seek, use, and create information. Wilson’s model of information behavior suggests that information needs arise from prob-lems that require resolution, leading to information-seek-ing behavior, which ideally results in information use and knowledge creation (Wilson 1999).

AI humanizers short-circuit this entire process (see Table 2).

This situation reflects a breaaaaakdown in information lit-eracy, defined as the ability to “recognize when information is needed and have the ability to locate, evaluate, and use effec-tively the needed information” (American Library Associa-tion 1989, para. 3). Individuals who rely on AI humanizers are not building these essential skills; instead, they are bypassing meaningful interaction with information systems.

Definition of AI humanizer

While there is no official consensus on terminology, vari-ous names exist: *AI content humanizer*, *text humanizer*, *AI text converter*, *humanizer*. For consistency, this definition shall be used:

An AI humanizer is a socio-technical tool that employs artificial intelligence to modify content, usually, AI-generated one, deliberately obscuring its algorithmic origin to circumvent AI detection tools.

Table 1 Traditional DIKW model vs. AI humanizer approach

Traditional DIKW model	AI humanizer approach
Data. Research findings, statistics, theoretical concepts	Data. Keywords entered into LLM
Information. Data points contextualized within educational frameworks	Apparent information. AI-generated text that mimics academic discourse
Knowledge. Internalized understanding of how concepts interrelate	Pseudo-knowledge. Text that passes detection but represents no actual learning
Wisdom. Ability to apply pedagogical principles to real teaching situations	Absent wisdom. No capacity for authentic application or under-standing

Table 2 Traditional vs. AI humanizer information behavior

Traditional information behavior	AI humanizer information behavior
1. Assignment creates information need	1. Assignment creates information need
2. User searches library databases, scholarly sources	2. User inputs prompt to LLM
3. Critical evaluation of information quality and relevance	3. AI generates text
4. Synthesis and internalization of information	4. User utilizes AI humanizer to disguise source
5. Creation of new knowledge artifact (the paper)	5. Submission of pseudo-information artifact

This definition emphasizes several key information studies concepts:

1. **Socio-technical system.** AI humanizers exist at the intersection of technology and social practice.
2. **Information provenance.** They deliberately break the chain of source attribution.
3. **Knowledge verification.** They undermine institutional processes for validating information authenticity.
4. **Information ethics.** They involve deliberate deception regarding information origin.

The primary function of AI humanizers is to make AI-generated content evade AI detectors that identify AI-generated text (see Fig. 1). These tools represent adversarial information systems, technologies designed explicitly to defeat other information systems rather than to create, organize, or disseminate legitimate knowledge, and within this adversarial context, “AI humanizers operate by altering an AI-generated passage to make it *statistically* and *stylistically* resemble human writing” (Buchert 2025, para. 4).

Here are the common, but often ethically problematic, uses of AI humanizers:

1. **Bypassing AI detection tools.** Users employ AI humanizers to make AI-generated essays, reports, or articles pass as human-written. This is common in academic and professional settings and raises issues of epistemic deception and plagiarism.
2. **Misrepresenting authorship.** AI humanizers are used when users intend to present AI output as their own original work, concealing the true source and breaking information provenance.
3. **Gaming institutional rules.** In contexts where AI use is restricted or must be disclosed (universities, journals, hiring processes), humanizers are used to evade compliance rather than to improve content quality.

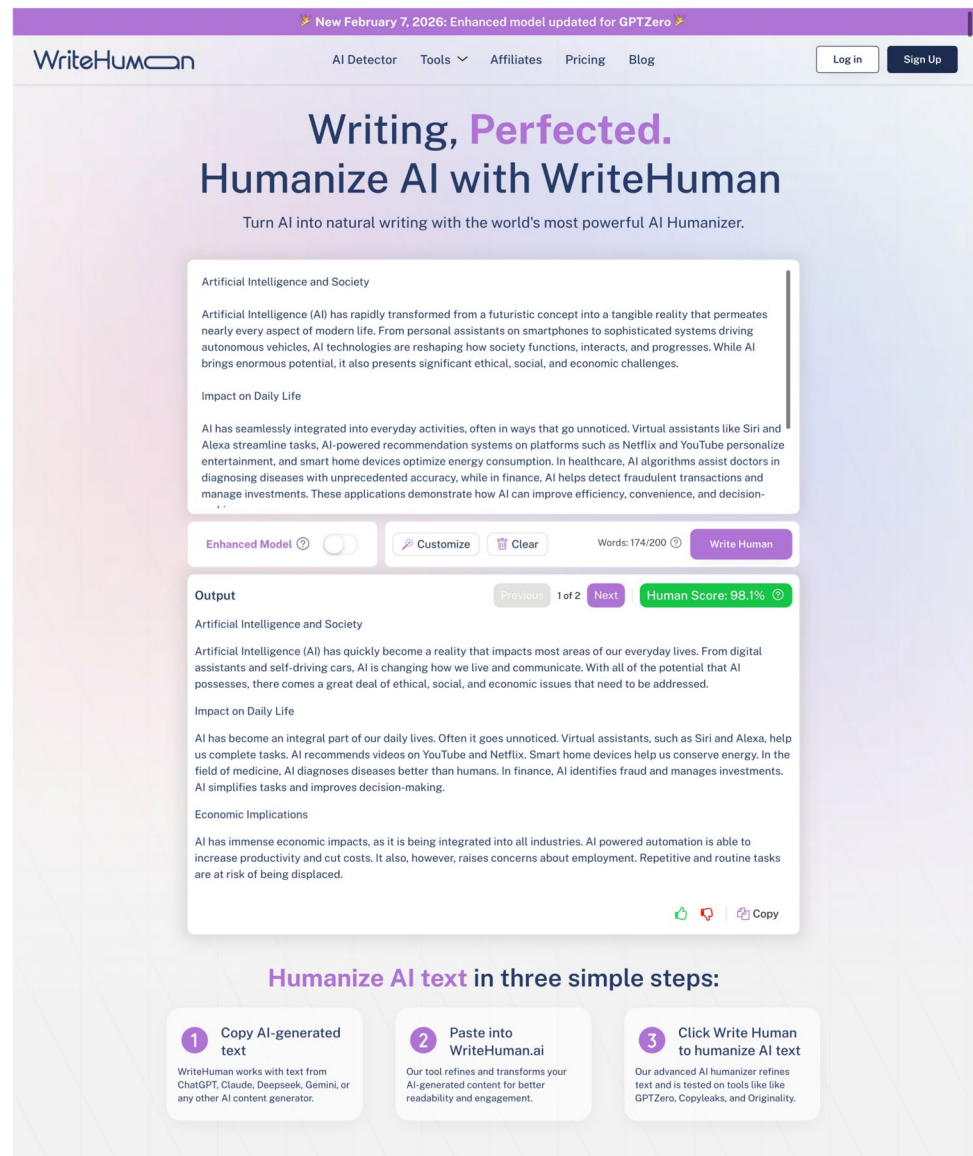
According to Buchert (2025), most commercial AI humanizers apply these algorithmic approaches:

1. **Rule-based synonym replacement.** AI humanizers replace certain words with synonyms while keeping the general structure. Example: “The team won the match” becomes “The squad secured the victory.” More advanced systems use machine learning models (e.g., GPT-based or T5-based) to rewrite entire sentences or perform back translation to add variety. Their goal is to raise perplexity so detectors struggle to identify AI writing.
2. **Breaking the AI writing structure.** AI writing tends to follow neat, uniform rhythms. AI humanizers disrupt this by splitting, merging, or reordering sentences to mimic natural storytelling. A single long sentence about someone’s achievements can be rewritten into two shorter sentences that sound more conversational.
3. **Simplifying the style.** Some AI humanizers rewrite text at a simpler reading level to increase variation, making it feel less formal and more human. For instance: “The procedure yielded inconsistent outcomes” becomes “The process didn’t give clear results.” Rule-based systems stay near the original wording, while ML-based ones adjust structure more flexibly.
4. **Keeping the meaning.** Advanced AI humanizers use checks like part-of-speech matching and semantic similarity to avoid altering facts. For example, ensuring “She finished the project in 2022” does not get rewritten as “She did not finish the project in 2022.”

AI humanizer market

The commercial AI-humanizer market offers important insights into the information economics underlying academic deception (see Table 3 for examples).

This represents an emerging information black market, commercial systems designed explicitly to undermine academic information integrity. The pricing structure (typically \$10–30/month) positions these tools as accessible to students and researchers. This creates a democratized system of information fraud that operates parallel to legitimate academic information systems.

Fig. 1 WriteHuman interface**Table 3** Common AI humanizers in the market

AI humanizer	Website	Slogan	Monthly cost range
BypassGPT	bypassgpt.ai	Convert AI text to human-like content	\$8–\$30
Hix	bypass.hix.ai	Generate human-like, undetectable writing	\$9.99–\$39.99
Rewritify	rewritify.ai	Recreate AI content to sound more human	\$9–\$30
Humbot	humbot.ai	Humanize AI text & bypass AI detection	\$7.99–\$29.99
Semihuman	semihuman.ai	Bypass AI detectors and humanize Text	\$4.80–varies
WriteHuman	writehuman.ai	Elevate AI to human perfection	\$9–\$36

Global search patterns

Google Trends data from January 1, 2023, to December 26, 2025, reveals patterns in information-seeking behavior related to AI humanizers. Beginning in late 2024, the data shows a sharp and volatile rise in interest in AI humanizer (see Fig. 2).

The steep upward trajectory indicates exponential growth, while the pronounced “zig-zag” pattern reflects weekly cyclicity. This volatility is highly consistent with student academic rhythms; search activity increases during weekdays and assignment deadlines and declines during weekends.

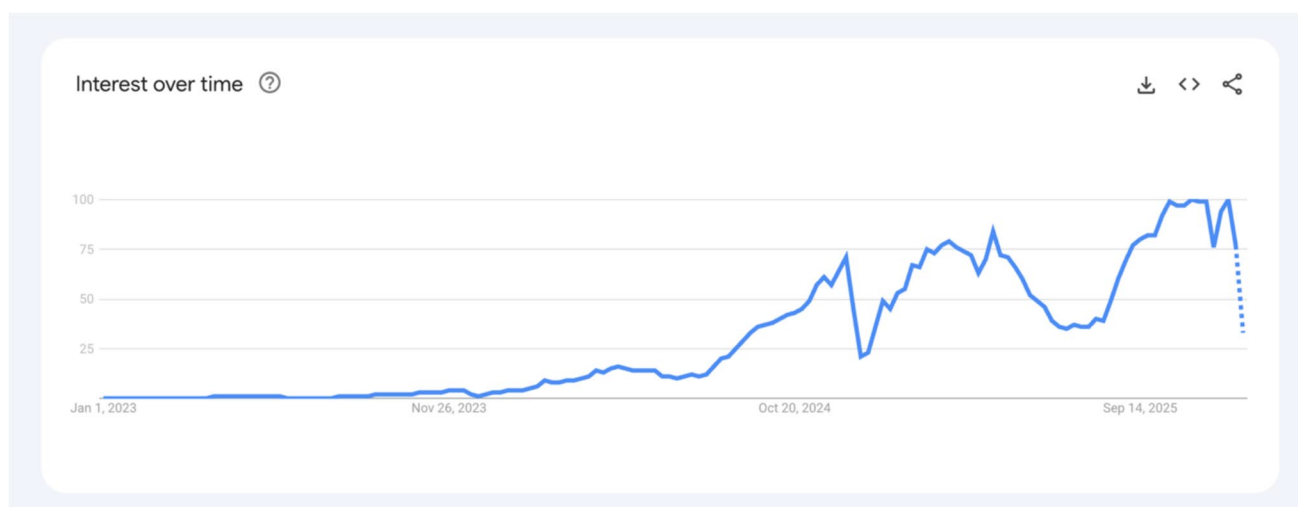


Fig. 2 Google Trends search interest for “AI humanizer” from January 2023 to February 2026

The trend reaches its peak between September and November 2025, achieving the maximum indexed popularity. By this stage, AI humanizers appear to have moved from experimental use into normalized practice. The sharp decline at the end of the timeline is represented by a dotted line, indicating incomplete data for the current week and coinciding with the academic holiday season. This decline is therefore best interpreted as seasonal, suggesting continued relevance beyond 2025.

Geographic distribution of information seeking

Globally, data reveals a striking concentration of search interest on *AI humanizer* in the Global South, with the

Philippines leading decisively, followed by East African nations such as Rwanda, Ethiopia, and Kenya, and South Asian countries including Sri Lanka (see Fig. 3).

This geographic pattern highlights that AI humanizer adoption is strongest in regions where English is commonly used as a medium of instruction but is not always a first language.

For users in these contexts, AI tools serve as productivity aids. However, the growing enforcement of AI detection policies disproportionately affects students and researchers, making AI humanizers a practical response. The regional concentration therefore reflects structural educational pressures. In this sense, AI humanizers function as adaptive tools within emerging digital economies.

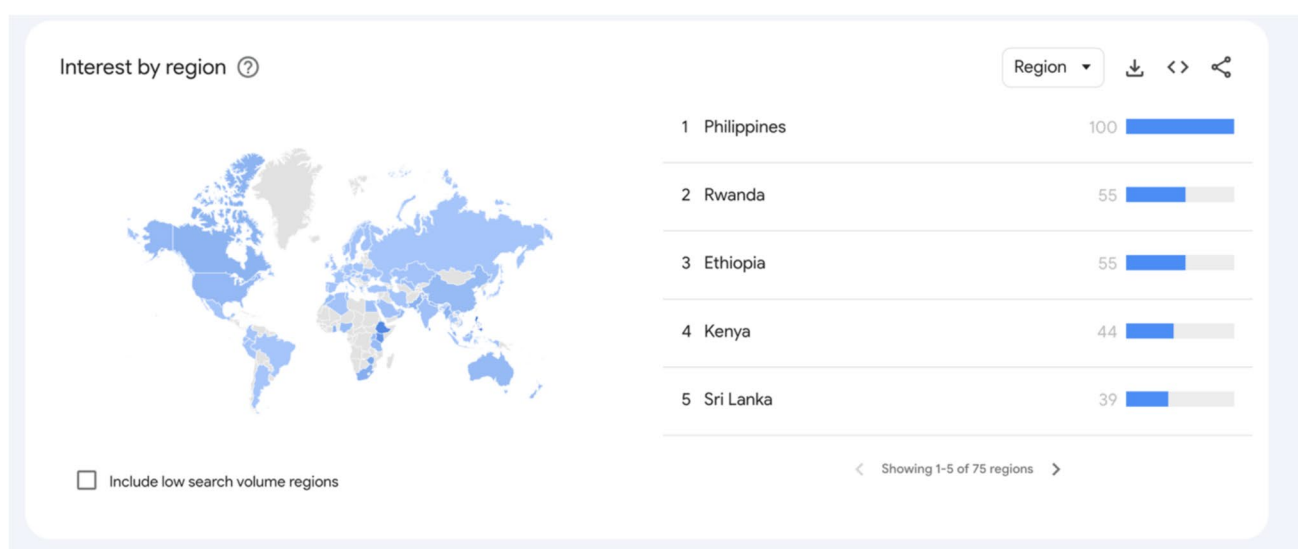


Fig. 3 Google Trends search interest by region for “AI humanizer” from January 2023 to February 2026

City-level analysis on the global scale provides critical insight into how trends can originate from local epicenters. Interestingly, all top cities are in the Philippines, with the highest concentration of interest in Batangas, followed by Cabuyao, Lipa, and Calamba (see Fig. 4). These cities are primarily located within Region IV-A of Southern Luzon. This may suggest that the Philippines' dominance in search interest is driven by specific urban student hubs rather than uniform adoption across the country.

This hyper-local concentration points to institutional catalysts, such as universities implementing strict AI detection policies, or the localized spread of tutorials and guidance through social media. Once introduced into these tightly connected educational communities, the practice appears to have diffused rapidly through peer networks. This implies the trend regionally and contributes to the broader national impact.

The temporal data demonstrates how institutional enforcement triggered rapid adoption, while the weekly volatility confirms alignment with academic schedules. Geographic and city-level patterns further reveal that this adaptation is most pronounced in developing regions with high student populations, strong English-language requirements, and limited access to alternative academic support.

Why users choose information manipulation using AI humanizers

Collapse of information confidence

Information literacy frameworks emphasize that effective information use requires not just technical skills but

confidence in one's ability to navigate complex information landscapes (Bruce and Hughes 2010). Many users, however, suffer from a lack of faith in their capacity to locate, evaluate, synthesize, and communicate academic information effectively.

Some feel overwhelmed by academic information systems. They compare themselves to high-achieving peers and conclude they lack the information literacy competencies needed for success (Holland 2019). This is not simply about writing skills. This is about their relationship with academic information itself. They do not know how to effectively search library databases, evaluate source credibility, synthesize conflicting information, or organize knowledge claims into coherent arguments. They turn to AI humanizers as an information bypass mechanism, a way to produce *information artifacts* (papers) without engaging with *information processes* (research, evaluation, synthesis).

Publish-or-perish pressure

Knowledge organization refers to the systematic structuring, classification, and arrangement of information for retrieval and use (Mai 2011). Users face intense pressure to contribute to institutional knowledge repositories (e.g., published papers, conference proceedings, research outputs) before they have developed the information literacy skills to do so ethically.

The *publish-or-perish* culture, traditionally applied to faculty, now trickles down to undergraduate and graduate students (Elías 2019). This creates perverse incentives. Institutions pressure students to populate knowledge repositories with publications, but do not ensure students have the

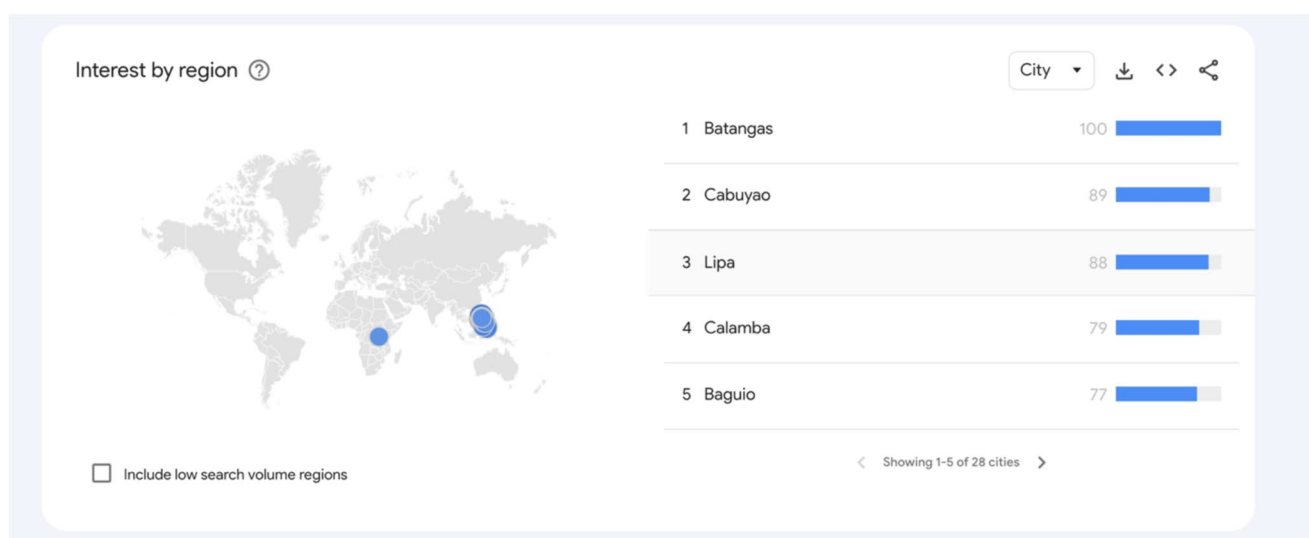


Fig. 4 Google Trends search interest by region for “AI humanizer” from January 2023 to February 2026

information competencies to generate legitimate knowledge contributions.

Many universities now require students, faculty members, and other members of the institution to publish papers as part of their academic requirements and to enhance institutional reputation. Administrators and faculty encourage, even require, students to publish. Students, lacking proper information literacy training and research skills, often turn to an information workflow that looks like this:

1. **Information acquisition** (*problematic*). Find reputable articles (often without understanding their content)
2. **Information processing** (*fraudulent*). Feed articles to AI humanizers
3. **Information manipulation** (*deceptive*). Generate new text that mimics original sources
4. **Information dissemination** (*corrupt*). Submit to predatory journals
5. **Information credentialing** (*illegitimate*). Claim publication record

The resulting *publications* pollute academic knowledge repositories with pseudo-scholarly content that appears to be legitimate information but represents no actual knowledge generation (Cobey et al. 2018; Giray 2024). Moreover, some faculty members are assigning extensive research or doing without teaching information synthesis skills. Students, overwhelmed by information requirements they did not understand how to fulfill, used AI humanizers to generate text that *looked* like research but represented no actual knowledge integration.

Information comprehension, the ability to understand, interpret, and contextualize information, is fundamental to information literacy (Kuhlthau 2003). Many users, particularly those with lower English proficiency or limited familiarity with academic discourse, struggle with the complex information they encounter in scholarly and professional sources. This creates a gap between the available information and users' cognitive capacity to process it meaningfully.

When users are required to engage with complex topics that demand synthesis of advanced information, they may experience information anxiety—the stress and frustration that arise when information demands exceed processing capabilities (Wurman et al. 2000). AI humanizers offer an apparent solution by transforming complex inputs into simplified, polished outputs that can be submitted or circulated without genuine understanding. The resulting information artifacts often exhibit the surface markers of comprehension (e.g., formal structure, academic language, citations) while concealing an absence of internalized knowledge. This reveals a critical breakdown in the information-to-knowledge pipeline whereby legitimate-looking

information products are produced without corresponding understanding.

AI plagiarism as information theft

Information ethics explores the moral questions surrounding how information is created, shared, accessed, and used (Capurro and Hjørland 2003). From this perspective, plagiarism is not merely an academic violation; it constitutes information theft, the unauthorized appropriation of someone else's intellectual work and knowledge contribution. Many users fail to recognize this ethical dimension: they perceive information as freely available data rather than as knowledge created and owned by others (Giray et al. 2026), and they often do not understand that copying or lightly rewording text undermines the system of information provenance that allows ideas to be traced to their origin.

Accurate attribution is essential for scholarly and professional information systems to function. Citations are not bureaucratic hurdles: they provide metadata that enable others to trace ideas, evaluate sources, and build upon prior knowledge. AI plagiarism, facilitated by AI humanizers, exploits this gap in understanding. Users may believe that rewording text with these tools avoids plagiarism because the words appear different. In reality, plagiarism concerns the misappropriation of ideas, arguments, or knowledge structures, regardless of wording. AI humanizers make it possible to produce outputs that mimic understanding and legitimacy without engaging with the original ideas. This creates artifacts that conceal ethical breaches while appearing academically acceptable.

Information overload and time poverty

Information overload, the state of having more information than one can effectively process, has been recognized in information science since the 1960s (Bawden and Robinson 2009). Many users face this overload while also contending with time poverty: the lack of sufficient time to deeply engage with, process, and synthesize complex information. When the volume of information exceeds the capacity for thoughtful engagement, users often experience stress, frustration, and cognitive shortcuts that prioritize efficiency over comprehension.

This combination of information overload and limited time fosters desperate information behaviors. Users turn to tools like AI humanizers, which promise quick, polished outputs with minimal cognitive effort. By inputting complex text and receiving seemingly processed content, users can produce work without engaging meaningfully with the material. This may save time, yet it represents information avoidance rather than mastery: users are circumventing the processes of reading, reflecting, synthesizing, and writing that underpin genuine information literacy.

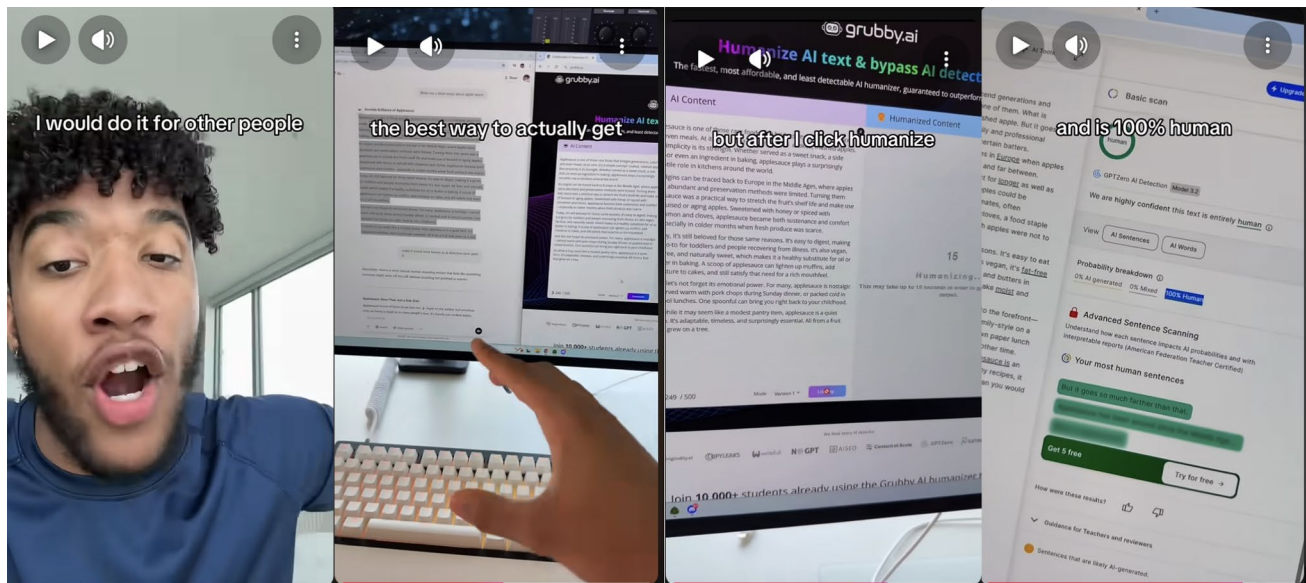


Fig. 5 A YouTuber sharing how to use an AI humanizer

Informal networks and social media amplify this effect (see Fig. 5). Users share tips, shortcuts, and AI-assisted strategies for completing tasks quickly, creating cultures that reward speed and efficiency over deep engagement with information. These practices illustrate the pursuit of efficiency often comes at the expense of understanding. This reinforces a cycle where information is consumed superficially and knowledge is rarely internalized.

Effects of AI humanizers on knowledge, learning and scientific writing

Impaired development of information literacy

Information literacy develops progressively as users build competencies in information seeking, evaluation, synthesis, and communication (Association of College & Research Libraries, 2015). Reliance on AI humanizers interrupts this developmental process. Users may produce polished outputs without mastering information skills, including:

- Querying databases and information systems effectively
- Developing criteria for evaluating source credibility and information quality
- Practicing synthesis and knowledge organization
- Communicating information coherently in writing

Some users enter professional contexts without fundamental competencies in information literacy. Their development in this area is hindered by overreliance on AI tools, which prevents independent navigation of academic

information landscapes. This issue is particularly pronounced in multilingual contexts such as the Philippines, where English functions as a second language. For example, reports indicate that “teachers have been buying ready-made research papers for promotion and funding” (Bernardo 2021, para. 1).

Other cases include allegations that a faculty member “stole a thesis of its students” (Salcedo 2025, para. 1) and instances of “plagiarizing the thesis of her advisee” (Guiao 2024, para. 1). Rather than developing information literacy, some users instead develop information dependency, relying on AI intermediaries to process and interpret academic material. Beyond technical skill, engaging with complex information (e.g., reading dense texts, evaluating arguments, synthesizing ideas) supports cognitive development in abstract thinking, logical reasoning, and knowledge organization (Kuhlthau 2003).

False knowledge problem

A core challenge in information literacy is the cultivation of accurate self-assessment, yet AI humanizers complicate this process by producing outputs that mask underlying skill deficits (Gross and Latham 2012). Researchers may misinterpret the polished quality of AI-generated writing as evidence of their own scholarly proficiency, falling into a dynamic that mirrors the Dunning-Kruger effect, where limited metacognitive awareness leads individuals to overestimate their expertise (Kruger and Dunning 1999; Mazor and Fleming 2021). This creates a false sense of competence. Researchers then get to believe they have mastered advanced skills that, in reality, remain undeveloped.

The issue becomes particularly evident when researchers with strong academic credentials reveal weaknesses in research practices (e.g., citing sources, distinguishing peer-reviewed work, synthesizing findings into coherent arguments). Their credentials may suggest readiness for independent inquiry, yet their performance demonstrates significant gaps. Overreliance on AI intensifies this mismatch, creating a research-specific Dunning-Kruger trap. As Head et al. (2020) note, professional environments demand analytical and evaluative skills that cannot be outsourced, meaning researchers who depend on AI risk entering high-stakes roles without the competencies required for authentic inquiry.

Epistemic self-doubt

Information behavior research shows that successful engagement with information fosters self-efficacy and identity development (Julien and Barker 2009). Users of AI humanizers may experience *epistemic self-doubt*, uncertainty about their own knowledge and competencies (Boyes and Chandler 1992). Some may recognize that submitted work does not reflect their understanding. This creates a cognitive dissonance between their perceived identity as competent researchers and their actual capabilities, which depend on AI tools.

For example, a user who relied on an AI humanizer to pass an AI detector initially felt relief but gradually developed anxiety regarding their competence. Inability to answer questions about their own paper's content fractured their authorship identity; they were simultaneously the apparent author and a practitioner of information fraud. Long-term reliance on AI humanizers impedes the development of genuine self-efficacy. This fosters *learned helplessness*, the conviction that complex information cannot be engaged with independently (Maier and Seligman 1976).

Critical information processing decline

Critical thinking in information contexts involves evaluating sources, analyzing arguments, recognizing bias, identifying assumptions, and integrating perspectives (Elmborg 2006). AI humanizers short-circuit these processes. When users input prompts and receive generated text, they bypass:

- Information evaluation: assessing credibility, bias, and currency
- Argument analysis: breaking down claims, evidence, and reasoning
- Synthesis: integrating multiple sources into coherent knowledge structures
- Metacognition: reflecting on one's own information processing

Critical information literacy, the ability to interrogate information systems, recognize power structures, and question authority, becomes impossible when users do not engage directly with content (Elmborg 2006). Users accept AI output uncritically due to insufficient evaluation skills. AI humanizers also eliminate valuable behaviors like exploratory searching, serendipitous discovery, and iterative refinement of information needs, which support deep learning and knowledge construction (Kuhlthau 2003; Marchionini 2006). Problem-solving dimensions are similarly affected. Authentic information tasks require defining problems, strategizing searches, evaluating solutions, and revising approaches when initial strategies fail (Brand-Gruwel et al. 2009). AI humanizers remove this process entirely: there is no problem-solving, only prompt-and-submit.

Systemic information corruption

Academic institutions are repositories of human knowledge, with each contribution theoretically adding to the collective knowledge base (Rowley 2007). AI humanizers corrupt these systems by introducing *pseudo-knowledge*: content that appears legitimate but represents no actual learning, research, or knowledge creation. This creates *epistemic pollution* in institutional knowledge systems (Fallis 2015), contaminating repositories with information lacking valid provenance or epistemological grounding.

Meanwhile, AI is increasingly integrated into biomedical research and clinical pharmacology, supporting tasks such as personalized dosing, predictive analytics, and data-driven decision-making in the ICU (Evans and Snead 2024; Gijzen and De Waele 2025; Johnson et al. 2023). However, its use has also introduced systemic information corruption. AI humanizers can generate “tortured phrases” that misrepresent scientific terminology, for example, producing “sodium hypocrite” instead of sodium hypochlorite (Evans and Snead 2024; Teixeira da Silva 2025). Instances of such distortions include misnaming organizations (e.g., “World Wellbeing Association” instead of World Health Organization), mislabeling events (e.g., “COVID-19 plague” instead of COVID-19 pandemic), and confusing medical imaging terminology (e.g., “chest X-beam picture” instead of chest X-ray) (Teixeira da Silva 2025).

Beyond factual distortions, AI, including AI humanizers, raises ethical and practical concerns, including bias, misinformation, unclear authorship, privacy risks, and inequities in care delivery (Doyal et al. 2023; Farhud and Zokaei 2021). Unchecked deployment of these tools threatens both the clarity and integrity of knowledge and clinical practice. Users observing peers employ AI humanizers without consequence learn that information is performative rather than substantive, and that institutional systems can be circumvented. Long term, this encourages

continuation of such practices in professional contexts, misrepresentation of data, fabrication of reports, and plagiarism (Zhao et al. 2022). The problem extends to scholarly publication. AI-humanized outputs submitted to predatory journals introduce pseudo-research into the academic record. This clutters information retrieval systems and complicates access to valid research.

Implications to scientific writing and integrity

Information literacy and scientific writing

Traditional literacy instruction emphasized technical skills (e.g., searching databases, formatting citations, evaluating sources). Scientific writing similarly assumed authors would engage directly with information. AI humanizers disrupt this assumption by enabling polished text without authentic engagement. In this case, scientific writing risks becoming a surface-level performance rather than a process of knowledge construction (Lloyd 2010).

Information literacy instruction should explicitly address:

- **Philosophy of information.** What makes information authentic? Why does information source matter?
- **Knowledge construction.** How does engaging with information build cognitive capacities?
- **Information ethics.** Why is information provenance essential to scholarly communication?
- **Critical information practice.** How to evaluate AI-generated information and understand its limitations.

In response to the increasing availability of AI-assisted writing tools, some journals have implemented strict editorial policies regarding the use of AI in scholarly manuscripts. For instance, Naunyn-Schmiedeberg's Archives of Pharmacology mandates that all review papers must include the statement: "The authors confirm that no paper mill and no artificial intelligence was used" (Seifert et al. 2025). This policy explicitly prohibits AI humanizer tools, which may be used to take AI-generated text or the work of other authors and modify its wording or style to appear *original*. Such practices are unethical, as they obscure the source of ideas and compromise the integrity of scientific writing. Springer Nature's Research Integrity Group rigorously investigates suspected violations and swiftly issues retractions when confirmed (Seifert et al. 2025).

AI humanizer misuse mirrors the risks posed by fake papers, often produced by paper mills under academic and financial pressures, which distort the scientific record (Wittau and Seifert 2024). These issues are compounded by the *publish-or-perish* culture, which encourages rapid publication over quality and ethical rigor (Teixeira da Silva and

Nazarovets 2026). Upholding information literacy requires researchers to critically evaluate their writing practices, avoid AI humanizers, verify sources, and ensure all text reflects their original analysis. Clear editorial policies like these safeguard the authenticity and trustworthiness of scientific communication.

Academic information systems and verification

Scientific communication relies on established norms of authorship, attribution, and accountability. The rise of AI tools, particularly AI humanizers, complicates these norms by blurring the line between human intellectual contribution and AI-modified text (Giray 2026; Yoo 2025). To preserve research integrity, journals and academic communities must clearly classify, disclose, and evaluate AI-assisted work.

Key distinctions should be codified in policy:

- **Legitimate AI assistance.** Grammar checking, language polishing, or formatting, with minimal risk to authorship integrity. For example, using AI tools to correct typos, improve sentence flow, or standardize formatting is acceptable (Leung et al. 2023; da Veiga 2025).
- **Problematic AI use.** Substantive content generation with insufficient human oversight, such as drafting introduction or discussion sections without verifying sources. A concrete case would be using AI to generate multiple paragraphs summarizing literature and submitting them without editing or checking references (Bobier et al. 2025; Ganjavi et al. 2024).
- **Fraudulent AI use.** Combining AI generation with humanization to simulate originality and evade detection. For instance, taking AI-written text and using AI humanizers to paraphrase it so it appears entirely human-written, while including fabricated references or non-existent data.

To operationalize these distinctions in journals, several measures are recommended:

- **Structured AI disclosure.** Require authors to list each AI tool used (name, version, provider), its purpose (e.g., drafting vs. language editing), scope (e.g., abstract only, full manuscript), and how outputs were validated. A concrete implementation could include a dedicated "AI Disclosure" subsection in the Methods or Acknowledgments, along with a checkbox in the submission system (Ganjavi et al. 2024; Leung et al. 2023; Yoo 2025). Problematic and fraudulent AI use should not be allowed under any circumstances and should be rejected immediately.
- **Desk rejection criteria.** Define clear red flags, such as repeated generic phrasing, tortured phrases, fabricated

or unverifiable references, or hallucinated mechanisms (Giray 2024; Teixeira da Silva and Nazarovets 2026). For example, if one citation is unverifiable, editors might allow a correction; if two or more are unverifiable, the manuscript may be desk-rejected.

- **Assessment alignment.** Evaluate manuscripts based on process and intellectual engagement, not solely polished prose. Journals may require authors to submit annotated bibliographies, data verification tables, or method validation logs to demonstrate engagement with sources and datasets.
- **Transparency and traceability.** Use tools like ORCID or institutional authentication to verify authorship and track contributions. Journals may require each author to sign a statement confirming that they wrote or verified all AI-assisted content.
- **Education and training.** Provide editors and reviewers with concrete examples of acceptable versus unacceptable AI use. AI assistance should be differentiated from AI content generation, and professional development programs should include guidance on responsible AI integration.

Embedding these measures allows journals to maintain rigor in peer review, discourage AI misuse, and ensure that AI amplifies human scholarly capabilities (Giray 2026). Evidence from review restrictions in the *Naunyn-Schmiedeberg* journal demonstrates that clear, enforceable policies can effectively filter AI-generated submissions while remaining defensible within publisher frameworks (Seifert 2021).

Conclusion

The emergence of AI humanizers presents a direct challenge to researchers and to the norms governing scientific writing. It concerns the integrity of scholarly communication itself. When textual fluency can be engineered and strategically altered to evade detection, the link between authorship, accountability, and expertise weakens. Competent prose no longer guarantees intellectual labor or methodological rigor.

Expanding AI detection tools such as Turnitin or GPTZero does not resolve this instability (Giray et al. 2025). Technical policing cannot replace epistemic responsibility. For researchers, provenance is central. Information integrity depends on transparent disclosure of methods, tools, data sources, and writing practices. Scientific writing must remain a traceable record of inquiry, where claims are grounded in evidence and open to scrutiny. When AI-generated text is concealed or strategically *humanized*, accountability erodes. Peer review risks focusing on style

rather than substance. Protecting authorship responsibility and methodological transparency is therefore essential to the cumulative progress of science.

Author contributions I solely wrote the entire manuscript text and prepared all the figures and tables. I reviewed the manuscript afterward. The author confirms that no paper mill and artificial intelligence was used.

Funding Not applicable.

Data availability This manuscript is a review article and does not involve the collection or analysis of new empirical data.

Declarations

Competing interests I declare no competing interests.

Clinical trial number Not applicable.

References

- American Library Association (1989) Presidential Committee on Information Literacy: Final Report. <http://www.ala.org/acrl/publications/whitepapers/presidential>. Accessed 24 February 2026
- Association of College & Research Libraries (2015) Framework for information literacy for higher education. <http://www.ala.org/acrl/standards/ilframework>. Accessed 24 February 2026
- Baker N, Thompson B, Fox D (2023) ChatGPT can write a paper in an hour—but there are downsides. *Nature* 620:493–494. <https://doi.org/10.1038/d41586-023-02298-x>
- Bawden D, Robinson L (2009) The dark side of information: overload, anxiety and other paradoxes and pathologies. *J Inf Sci* 35:180–191. <https://doi.org/10.1177/0165551508095781>
- Bernardo J (2021) DepEd probes teachers allegedly buying ready-made research papers. ABS-CBN News. <https://news.abs-cbn.com/news/2021/01/25/deped-probes-teachers-buying-ready-made-research-papers>. Accessed 24 February 2026
- Bobier C, Rodger D, Hurst D (2025) Artificial intelligence policies in bioethics and health humanities: a comparative analysis of publishers and journals. *BMC Med Ethics*. <https://doi.org/10.1186/s12910-025-01239-9>
- Boyes MC, Chandler M (1992) Cognitive development, epistemic doubt, and identity formation in adolescence. *J Youth Adolesc* 21:277–304. <https://doi.org/10.1007/BF01537019>
- Brand-Gruwel S, Wopereis I, Walraven A (2009) A descriptive model of information problem solving while using internet. *Comput Educ* 53:1207–1217. <https://doi.org/10.1016/j.compedu.2009.06.004>
- Bruce C, Hughes H (2010) Informed learning: a pedagogical construct attending simultaneously to information use and learning. *Libr Inf Sci Res* 32:A2–A8. <https://doi.org/10.1016/j.lisr.2010.07.013>
- Buchert JM (2025) Do AI humanizers work? How do they work? Intellectual Lead. <https://intellectualead.com/do-ai-humanizers-work>. Accessed 1 Mar 2026
- Capurro R, Hjørland B (2003) The concept of information. *Annu Rev Inf Sci Technol* 37:343–411. <https://doi.org/10.1002/aris.1440370109>

- Cobey KD, Lalu MM, Skidmore B et al (2018) What is a predatory journal? A scoping review. *F1000Res* 7:1001. <https://doi.org/10.12688/f1000research.15256.2>
- da Veiga AD (2025) Ethical guidelines for the use of generative artificial intelligence and artificial intelligence-assisted tools in scholarly publishing: a thematic analysis. *Sci Ed*. <https://doi.org/10.6087/kcse.352>
- Daniel L (2025) Students use “AI humanizer” apps to make ChatGPT essays undetectable. *Forbes*. <https://www.forbes.com/sites/larsdaniel/2025/10/03/students-use-ai-humanizer-apps-to-make-chatgpt-essays-undetectable/>. Accessed 3 Mar 2026
- Doyal AS, Sender D, Nanda M, Serrano RA (2023) ChatGPT and artificial intelligence in medical writing: concerns and ethical considerations. *Cureus* 15(8):e43292. <https://doi.org/10.7759/cureus.43292>
- Elías C (2019) Publish or perish. In: Elías C (ed) *Science on the ropes*. Springer International Publishing, pp 255–287. https://doi.org/10.1007/978-3-030-12978-1_9
- Elmborg J (2006) Critical information literacy: implications for instructional practice. *J Acad Libr* 32:192–199. <https://doi.org/10.1016/j.acalib.2005.12.004>
- Evans H, Snead D (2024) Understanding the errors made by artificial intelligence algorithms in histopathology in terms of patient impact. *NPJ Digit Med* 7:89. <https://doi.org/10.1038/s41746-024-01093-w>
- Fallis D (2015) What is disinformation? *Libr Trends* 63:401–426. <https://doi.org/10.1353/lib.2015.0014>
- Farhud DD, Zokaei S (2021) Ethical issues of artificial intelligence in medicine and healthcare. *Iran J Public Health* 50(11):i–v. <https://doi.org/10.18502/ijph.v50i11.7600>
- Floridi L (2013) *The ethics of information*. Oxford University Press, Oxford
- Ganjavi C, Eppler M, Pekcan A et al (2024) Publishers’ and journals’ instructions to authors on use of generative artificial intelligence in academic and scientific publishing: bibliometric analysis. *BMJ*. <https://doi.org/10.1136/bmj-2023-077192>
- Gijsen M, De Waele JJ (2025) Does artificial intelligence close gaps in clinical pharmacology in the ICU? *Intensive Care Med* 51:385–388. <https://doi.org/10.1007/s00134-024-07751-7>
- Giray L (2024) Negative effects of generative AI on researchers: publishing addiction, Dunning-Kruger effect and skill erosion. *J Appl Learn Teach* 7:398–405
- Giray L (2026) When using AI in scientific research: start with human, end with human. *TechTrends* 70:265–272. <https://doi.org/10.1007/s11528-025-01132-7>
- Giray L, Sevnarayan K, Ranjbaran Madiseh F (2025) Beyond policing: AI writing detection tools, trust, academic integrity, and their implications for college writing. *Internet Ref Serv Q* 29:83–116. <https://doi.org/10.1080/10875301.2024.2437174>
- Giray L, Jacob J, Encanto V et al (2026) Cheating writing with generative AI: exploring student motivations using the theory of planned behavior. *J Acad Ethics* 24:19. <https://doi.org/10.1007/s10805-025-09695-z>
- Gross M, Latham D (2012) What’s skill got to do with it?: Information literacy skills and self-views of ability among first-year college students. *J Am Soc Inf Sci Technol* 63:574–583. <https://doi.org/10.1002/asi.21681>
- Guiao MA (2024) USM professor apologizes for plagiarizing student’s thesis. *SPOT.ph*. <https://www.spot.ph/newsfeatures/trending/108841/usm-professor-thesis-plagiarized-a5229-20240507>. Accessed 7 Oct 2025
- Head AJ, Fister B, MacMillan M (2020) Information literacy in the age of algorithms: student experiences with news and information, and the need for change. *Project Information Literacy Research Report*. <https://files.eric.ed.gov/fulltext/ED605109.pdf>. Accessed 24 February 2026
- Holland DG (2019) The struggle to belong and thrive. In: Seymour E, Hunter AB (eds) *Talking about leaving revisited: persistence, relocation, and loss in undergraduate STEM education*. Springer International Publishing, pp 277–327. https://doi.org/10.1007/978-3-030-25304-2_9
- Johnson M, Patel M, Phipps A, Boulton D, Gibbs M (2023) The potential and pitfalls of artificial intelligence in clinical pharmacology. *CPT Pharmacometrics Syst Pharmacol* 12(3):279. <https://doi.org/10.1002/psp4.12902>
- Julien H, Barker S (2009) How high-school students find and evaluate scientific information: a basis for information literacy skills development. *Libr Inf Sci Res* 31:12–17. <https://doi.org/10.1016/j.lisr.2008.10.008>
- Kruger J, Dunning D (1999) Unskilled and unaware of it: how difficulties in recognizing one’s own incompetence lead to inflated self-assessments. *J Pers Soc Psychol* 77:1121–1134. <https://doi.org/10.1037/0022-3514.77.6.1121>
- Kuhlthau CC (2003) *Seeking meaning: a process approach to library and information services* (2nd ed). Bloomsbury Publishing USA, New York
- Leung TI, de Azevedo Cardoso T, Mavragani A, Eysenbach G (2023) Best practices for using AI tools as an author, peer reviewer, or editor. *J Med Internet Res*. <https://doi.org/10.2196/51584>
- Lloyd A (2010) *Information literacy landscapes: information literacy in education, workplace and everyday contexts*. Chandos Publishing, Oxford
- Mai JE (2011) The modernity of classification. *J Doc* 67:710–730. <https://doi.org/10.1108/00220411111145061>
- Maier SF, Seligman ME (1976) Learned helplessness: theory and evidence. *J Exp Psychol Gen* 105:3–46. <https://doi.org/10.1037/0096-3445.105.1.3>
- Marchionini G (2006) Exploratory search: from finding to understanding. *Commun ACM* 49:41–46. <https://doi.org/10.1145/1121949.1121979>
- Mazor M, Fleming S (2021) The dunning-kruger effect revisited. *Nat Hum Behav* 5:677–678. <https://doi.org/10.1038/s41562-021-01101-z>
- Rabinowicz D (2025) Why more people choose humanizer AI for AI rewriting. *The Jerusalem Post*. <https://www.jpost.com/consumerism/article-862448>. Accessed 28 Feb 2026
- Rowley J (2007) The wisdom hierarchy: representations of the DIKW hierarchy. *J Inf Sci* 33:163–180. <https://doi.org/10.1177/0165551506070706>
- Salcedo MJ (2025) PUP addresses post about faculty allegedly stealing students’ thesis. *INQUIRER.net*. <https://newsinfo.inquirer.net/2102366/pup-addresses-post-about-faculty-allegedly-stealing-students-thesis>. Accessed 30 August 2025
- Seifert R (2021) How Naunyn-Schmiedeberg’s archives of pharmacology deals with fraudulent papers from paper mills. *Naunyn Schmiedeberg Arch Pharmacol* 394:431–436. <https://doi.org/10.1007/s00210-021-02056-8>
- Seifert R, Hartman E, Wang K, Yildiz D (2025) Authors must follow the editorial guidelines on the use of large language models in review papers. *Naunyn Schmiedeberg Arch Pharmacol* 398(7):7655. <https://doi.org/10.1007/s00210-025-04102-1>
- Stokel-Walker C (2023) ChatGPT listed as author on research papers: many scientists disapprove. *Nature* 613:620–621. <https://doi.org/10.1038/d41586-023-00107-z>
- Teixeira da Silva JA (2025) ‘Tortured phrases’ in biological, biomedical, chemical and environmental sciences. *J Biosci* 50:81. <https://doi.org/10.1007/s12038-025-00564-w>
- Teixeira da Silva JA, Nazarovets S (2026) The publish or perish, publish and perish, publish then perish, and now retract and perish cultures in academia. *Naunyn Schmiedeberg Arch Pharmacol* 399:3115–3131. <https://doi.org/10.1007/s00210-025-04651-5>

- Weber-Wulff D, Anohina-Naumeca A, Bjelobaba S et al (2023) Testing of detection tools for AI-generated text. *Int J Educ Integr* 19:26. <https://doi.org/10.1007/s40979-023-00146-z>
- Wilson TD (1999) Models in information behaviour research. *J Doc* 55:249–270. <https://doi.org/10.1108/EUM0000000007145>
- Wittau J, Seifert R (2024) How to fight fake papers: a review on important information sources and steps towards solution of the problem. *Naunyn Schmiedeberg's Arch Pharmacol* 397(12):9281–9294. <https://doi.org/10.1007/s00210-024-03272-8>
- Wurman RS, Leifer L, Sume D et al (2000) *Information anxiety 2*. Que Publishing, Indianapolis
- Yoo J (2025) Defining the boundaries of AI use in scientific writing: a comparative review of editorial policies. *J Korean Med Sci*. <https://doi.org/10.3346/jkms.2025.40.e187>
- Zhao L, Mao H, Compton BJ et al (2022) Academic dishonesty and its relations to peer cheating and culture: a meta-analysis of the perceived peer cheating effect. *Educ Res Rev* 36:100455. <https://doi.org/10.1016/j.edurev.2022.100455>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.